

Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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Outline

1. Motivation & measurement — defects in CFRP, infrared stress (DSPSS)
2. Method — difference-normalized input & a GAT on the FEM mesh
3. Results — 19-class 3-D localization (macro-F1 61%)
4. New: robustness to measurement noise (defect-free negatives)
5. Outlook

Contributions

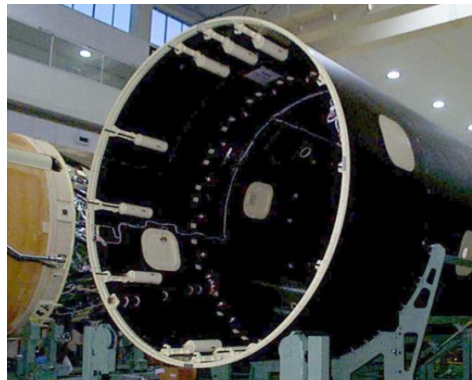
1. A **graph attention network** localizing 3-D delamination in *perforated* CFRP interstages directly from FEM surface stress (DSPSS).
2. **61% macro-F1** over 19 classes (region $\times$ layer) from *surface data only*, with defect false-negative rate $< 3\%$.
3. **Noise-robust** training (structured noise + defect-free negatives) — a step toward experimental infrared-stress data.

Introduction: CFRP in space transportation

CFRP carries primary load in reusable launch vehicles (high specific strength) [1, 2].

Interlaminar delamination around bolt holes critically degrades integrity, especially under repeated use. Reliable *defect localization* is essential.

Conventional NDT — ultrasonic [3], X-ray CT [4], tap testing [5] — is costly and operator-dependent [6].



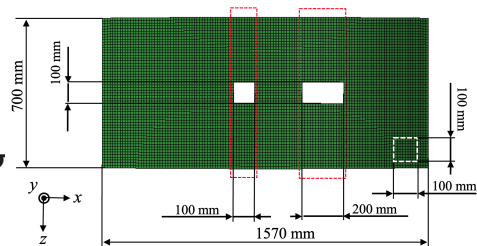
CFRP interstage structure of a launch vehicle.

Infrared stress measurement (DSPSS)

Thermoelastic effect relates a small temperature change to the change in the sum of principal stresses:

$$\Delta T = -k T \Delta \sigma_{\text{sum}}, \quad \sigma_{\text{sum}} = \sigma_1 + \sigma_2 + \sigma_3 = \text{tr}(\sigma)$$

Infrared stress measurement [7–9] thus yields a **full-field surface map** of σ_{sum} (DSPSS), from which sub-surface defects are inferred.



DSPSS shows a stress valley over the defect.

Previous study and purpose

Prior work. Kojima *et al.* [10, 11] — a CNN *proof-of-concept* on *non-perforated* CFRP from FEM DSPSS.

This work — harder setting: *perforated* interstage $\times$ 3-D location (region $\times$ layer) $\times$ extreme imbalance ($< 3\%$ defect nodes).

Problem — node classification on the mesh graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node features $\mathbf{x}_i = (x, y, z, \hat{\sigma}_{\text{sum}})_i$; learn $f_\theta : \mathcal{G} \rightarrow \mathbb{R}^{|\mathcal{V}| \times 19}$,

$$\hat{y}_i = \arg \max_{c \in \{0, \dots, 18\}} [\text{softmax } f_\theta(\mathcal{G})]_{i,c}, \quad \theta^\star = \arg \min_{\theta} \sum_{i \in \mathcal{V}} \text{FL}(p_{i,y_i}),$$

class 0 = defect-free; classes 1–18 = (in-plane region $\times$ insertion layer).

Related work: where this work sits

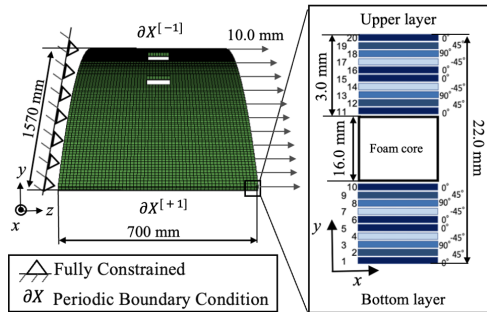
	Kojima '25 (Compos. B)	PIGMID '26 (Compos. B)	This work
Input	surface stress (FEM→IR)	guided waves (PZT×8)	full-field stress (DSPSS)
Model	transfer CNN	physics GNN (wave energy)	mesh-physics GNN
Geometry	simple coupons	instrumented plate	perforated inter- stage, 3-D
Physics	noise-aug. sim pretrain	wave-energy adjacency	FE-mesh topol- ogy + DSPSS
Novelty	sim-to-real transfer	path hetero- geneity	OOD gen. + de- tect/localize split

Unlike guided-wave GNNs (PIGMID) or simple-coupon transfer CNNs (Kojima '25), we learn on the structural FE mesh from full-field DSPSS, and are the first to quantify OOD size/position generalization — **detection transfers, precise localization collapses.**

FEM analysis conditions

- 1/80-scale curved panel ($R \approx 2$ m, $L \approx 1.57$ m, $t = 22$ mm).
- Holes of 100×100 and 200×100 mm.
- Sandwich laminate; tensile loading.
- Interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies.

Dataset: 1 reference + **1386** defect specimens; 5-fold CV (≈ 1110 train / 278 test per fold).



FEM boundary conditions & hole geometry.

Input data: normalized / difference DSPSS

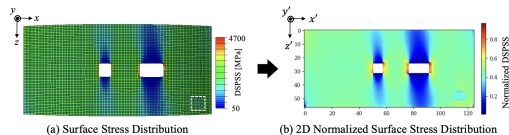
Raw DSPSS is dominated by **stress concentration around the hole**, masking the defect signal.

Remedy (paper):

- **Difference** field: damaged – defect-free reference;
- **Z-score normalization** (per node), improving depth discrimination.

$$d_i = \sigma_i^{\text{dmg}} - \sigma_i^{\text{ref}}, \quad \hat{\sigma}_i = \frac{d_i - \mu_d}{s_d}.$$

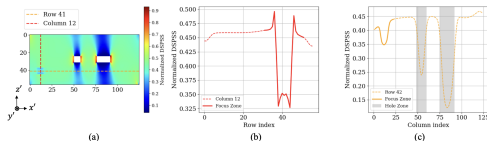
⇒ robust localization even near the opening.



Normalized DSPSS isolates the defect.

Stress distribution around the defect

- DSPSS exhibits a clear **stress valley** at the defect centre.
- Depth and width of the valley depend on **ply angle**: 90° plies give sharp narrow valleys; $0^\circ/45^\circ$ broaden them.
- This signature tells the network *which layer* the delamination sits in.



Normalized DSPSS profile across the defect.

Implication

Layer discrimination is a *fine spatial-frequency* problem — motivating local attention.

Detect first, then localize — screening comes for free

Two different jobs. *Detection* (is there / where?) needs **no learning**: every specimen shares one mesh, so healthy fields are a point cloud around *one* template.

The *same* per-node z-score we feed the GNN doubles as an unsupervised screen:

$$s_i = \frac{|x_i - \mu_i|}{\sigma_i}, \quad (\mu_i, \sigma_i) \text{ from } N=2000 \text{ hec}$$

Node-level, label-free, ~30 lines,
CPU-minutes.

Node AUROC (clean)

Per-node Mahalanobis / PCA:
0.999 / 0.995 (2×2 / 4×4).

Learned baselines: PaDiM 0.96, PatchCore 0.93, STFPM 0.75, diffusion 0.50–0.66.

SNR: robust, not fragile

Verified on a labelled coupon: detection holds (AUROC ~0.84) at noise $\sigma=0.1$ of field std, failing only near $\sigma \approx 0.7\text{--}1.0$ — not the folklore “0.1 collapse.” Lock-in IR ($\sigma \propto 1/\sqrt{K}$) adds margin.

⇒ detection is cheap; the GNN is for the hard part — **semantic 19-class (region ×**

Graph representation & GAT model

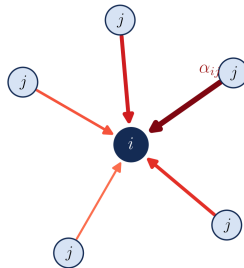
Nodes = elements ($\mathbf{x}_i$), edges = fixed mesh connectivity. GAT message passing [12–14]:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]),$$

$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_{k \in \mathcal{N}(i)} \exp e_{ik}},$$
$$\mathbf{h}'_i = \left\|_{m=1}^M \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j \right).$$

3 layers $64 \rightarrow 256 \rightarrow 256$, M heads, BN + dropout + residual.

GAT: attention-weighted aggregation over neighbours



Attention-weighted aggregation; arrow thickness $\propto \alpha_{ij}$.

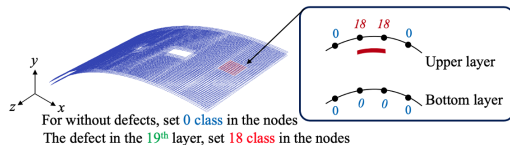
Output: 3-D defect location as 19-class node labels

- Node classification into **19 classes**:
class 0 = defect-free; remaining classes encode *in-plane region* $\times$ *insertion layer*.
- One-side labelling so prediction needs only single-surface DSPSS.

$$p_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=0}^{18} e^{z_{i,k}}}, \quad \hat{y}_i = \arg \max_c p_{i,c}.$$

Severe imbalance

Defect nodes are < 3% of all nodes —
handled by the loss (next slide).

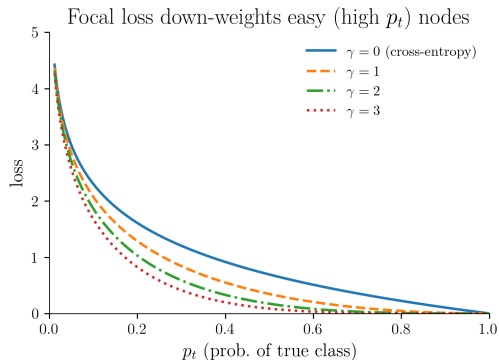


19-class label map.

Loss function: Focal loss

$$\text{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log p_t.$$

- Down-weights easy, abundant defect-free nodes; emphasises rare defect-class nodes [15].
- Crucial because intact regions dominate (> 97%).



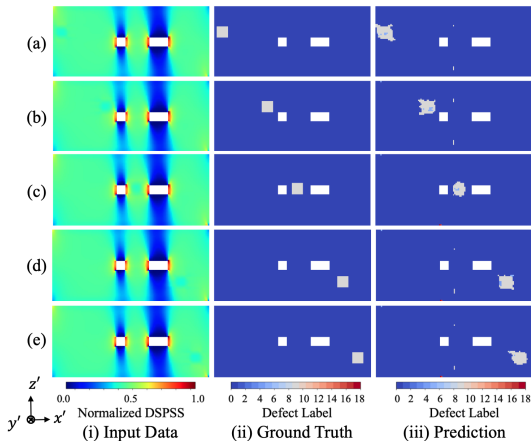
Results: prediction accuracy

Test-set metrics:

- planar $R = 0.72$ (mean), depth $P_{\text{gauss}} = 0.69$, combined TDPS = 0.70 (best 0.92).
- Best layers: 9–12 (mid-thickness); Layer-10 samples all TDPS ≥ 0.82 .
- False-negative rate for defect nodes $< 3\%$.

R : planar F1. P_{gauss} : Gaussian depth score ($\sigma=2$).

$$\text{TDPS} = \frac{1}{2}(R + P_{\text{gauss}}).$$



Input $\rightarrow$ ground truth $\rightarrow$ prediction (Layer-10 cases).

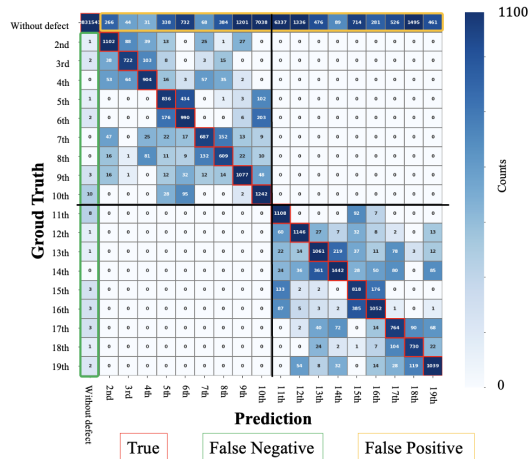
Overall performance & confusion

Headline

Macro-F1 = 61% across 19 classes,
using *surface DSPSS only*.

$$F1_c = \frac{2P_c R_c}{P_c + R_c}, \quad \text{macro-F1} = \frac{1}{19} \sum_{c=0}^{18} F1_c.$$

- Dominant diagonal; residual *adjacent-layer* confusion.
- False positives cluster near the hole edge.



Confusion matrix (macro-F1 61%).

Conclusion

- Introduced an **FEM + GAT** framework for 3-D delamination localization in perforated CFRP interstage panels.
- **Difference-normalized DSPSS** suppresses hole stress concentration and enables full-region estimation.
- Achieved **macro-F1 61%** on 19 classes from surface data only, with a very low defect false-negative rate and no false detections in defect-free regions.

A scalable foundation

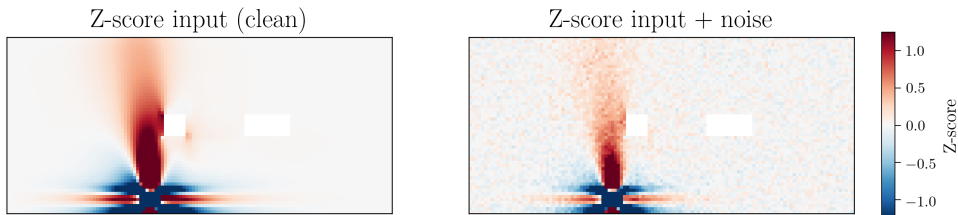
toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

New focus: robustness for real operation

The published model assumes clean FEM fields. In practice infrared measurement carries **acquisition noise**, and almost the whole structure is **defect-free** — false alarms are costly.

WCCM contribution

Robustness under realistic noise *and* explicit suppression of false detection on healthy regions.

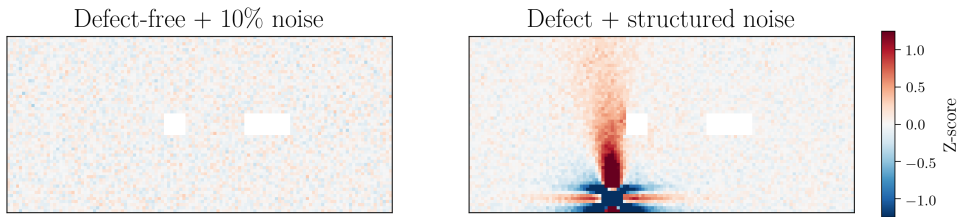


Clean vs. noise-injected Z-score input — the defect signature survives.

Noise-augmented dataset & defect-free negatives (NDF)

Defect-free negatives (NDF): zero field + Gaussian noise (10% of data std), 5,000 samples added to training.

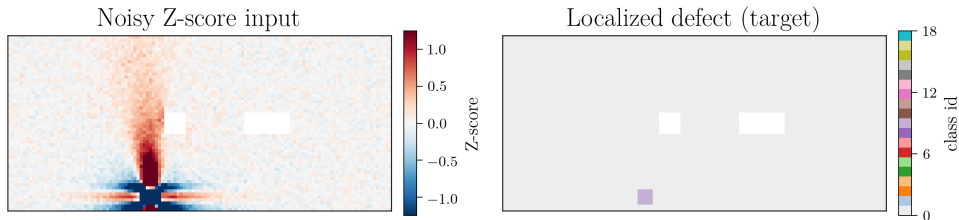
Structured noise on defect data: line noise along the fibre (z), attenuated noise in x , and global white noise.



Left: defect-free + noise (teaches “no defect”). Right: defect + structured noise.

Results under noise

- Difference-DSPSS input enables **full-region** defect estimation; prediction stays accurate on **noise-injected** data.
- **No false detection** on defect-free inputs — the NDF negatives remove spurious activations. $\Rightarrow$ robust enough to move toward *experimental* infrared stress data.



Noisy input (left) $\rightarrow$ correct defect localization (right).

Outlook [ongoing, beyond the published paper]

- **Sim-to-real:** apply the noise-robust model to *experimental* infrared-stress (TSA) measurements.
- **Architecture study (preliminary):** extending from attention GNNs to *mesh-physics* GNNs (MeshGraphNet) and layered variants; early results are encouraging and under evaluation with co-authors.
- **Application:** transfer to other perforated / bonded CFRP structures — rocket motor cases, aircraft wing boxes, satellite panels, payload-fairing SHM.

Preliminary; not part of the published results.

Acknowledgements

- Conducted in collaboration with **JAXA** (Research & Development Directorate, Research Unit IV).
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- Built on the FEA-generated CFRP interstage dataset of the prior study [10].

Thank you for your attention — questions welcome.

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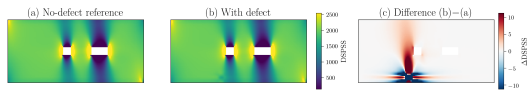
Appendix

Appendix: difference-data construction

Raw DSPSS near a hole is dominated by geometric stress concentration. We remove it by a per-specimen **difference**:

$$\text{Difference} = \underbrace{\text{Original (defect)}}_{\text{damaged}} - \underbrace{\text{Reference (no defect)}}_{\text{pristine}}.$$

The residual isolates the defect-induced perturbation, independent of the hole's baseline field.

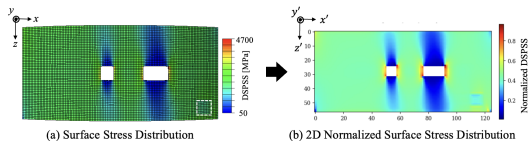


With-defect – no-defect reference = difference field (defect localized).

Appendix: Z-score normalization

The difference field is standardized per node (Z-score):

$$\hat{\sigma} = \frac{\sigma - \mu}{s}.$$

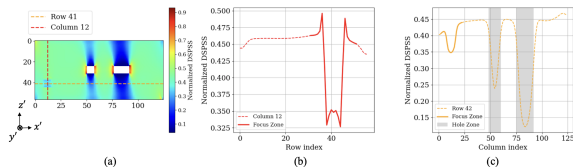


Normalized DSPSS input (paper figure).

- equalizes magnitude across regions/specimens;
- improves **depth-direction** (layer) discrimination;
- keeps the network NDT-admissible (coordinate/stress only).

Appendix: DSPSS profile around the defect

The difference-DSPSS shows a pronounced **stress valley** at the defect centre; its depth and width encode the insertion layer (ply-angle dependence).



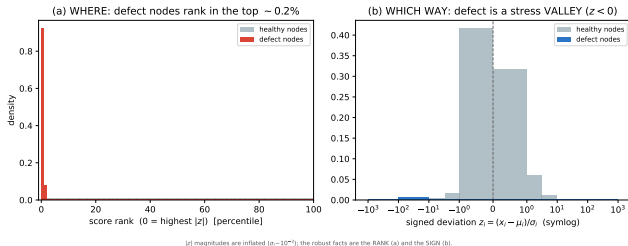
Profile of the normalized DSPSS across the defect (paper figure).

Appendix: evaluation metrics

- R — in-plane (planar) F1 score for defect vs. non-defect at the correct location.
- P_{gauss} — depth-direction score: Gaussian-weighted ($\sigma = 2$ layers) credit for predicting the correct or a near-correct insertion layer.
- **TDPS** (Total Defect Prediction Score) = $\frac{1}{2} (R + P_{\text{gauss}})$.
- **macro-F1** — unweighted mean F1 over all 19 classes (the headline metric).

These quantify in-plane localization (R), depth/layer accuracy (P_{gauss}), and their combination (TDPS), complementing the class-level macro-F1.

Appendix: what the Stage-0 screen keys on (one case)



Real 4×4 specimens, per-node

$$s_i = |x_i - \mu_i| / \sigma_i$$

- **Where:** defect nodes rank **top ~0.2%**; per-case **AUROC 0.998** (200 specimens).
- **Which way:** **98–100%** **negative** deviation $\Rightarrow$ a *stress valley* (thermoelastic signature).

Honest limits

$|z|$ magnitudes inflated ($\sigma_i \sim 10^{-3}$) — quote rank/sign, not “ $N\sigma$ ”. precision@K < 1: the hole edge also deviates $\Rightarrow$ the reason we *difference-normalize* before precise localization